

A Personalized Communication Application for Children with Autism through a Participatory Design Approach

Md. Tamjid Hosain*
Sami Uddin*

tamjid.hosain@northsouth.edu
sami.uddin02@northsouth.edu
Design Inclusion & Access Lab, North
South University
Dhaka, Bangladesh

Farhan Ahmed Fahim
farhan.fahim01@northsouth.edu
Design Inclusion & Access Lab, North
South University
Dhaka, Bangladesh

Mayisha Muntaha Alam
mayisha.alam@northsouth.edu
Design Inclusion & Access Lab, North
South University
Dhaka, Bangladesh

Shams Akbar Aalok
shams.aalok@northsouth.edu
Design Inclusion & Access Lab, North
South University
Dhaka, Bangladesh

Obayed Ur Rahman
obayed.rahman@northsouth.edu
Design Inclusion & Access Lab, North
South University
Dhaka, Bangladesh

Ashfaq Mahee Siddiky
ashfaq.siddiky@northsouth.edu
Design Inclusion & Access Lab, North
South University
Dhaka, Bangladesh

Marzan Mahatab
marzan.mahatab@northsouth.edu
Design Inclusion & Access Lab, North
South University
Dhaka, Bangladesh

Anik Sinha
anik.sinha@northsouth.edu
Design Inclusion & Access Lab, North
South University
Dhaka, Bangladesh

Nova Ahmed
nova.ahmed@northsouth.edu
Design Inclusion & Access Lab, North
South University
Dhaka, Bangladesh



Interactive board design



Final iteration of interactive board



Presenting interactive board and application to participants

Figure 1: Participatory design session II with interactive board and the application.

Abstract

Developing regions lack personalizable mobile applications with cultural context for CWA to aid in communication with parents and

*Both authors contributed equally to this research.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
CSCW Companion '24, November 9–13, 2024, San Jose, Costa Rica.
© 2024 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 979-8-4007-1114-5/24/11
<https://doi.org/10.1145/3678884.3681900>

teachers. This led us to design and develop an application to address this underexplored area. Three participants (parents and teachers of CWA) participated in two participatory design sessions to develop and improve the application. The same participants shared their views on the design and features of the application in the first study and provided feedback about them in the second study. There exists a literature gap on cultural context-based applications and a lack of personalization features for CWA. Our study aims to provide meaningful communication between CWA and others by developing a communication application that can be customized and personalized to cater to their needs. This work provides contextual design factors to HCI researchers by prototyping and developing a communication platform for CWA.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**.

Keywords

ASD, Autism, Communication, Participatory Design, South Asian Context

ACM Reference Format:

Md. Tamjid Hosain, Sami Uddin, Farhan Ahmed Fahim, Mayisha Muntaha Alam, Shams Akbar Aalok, Obayed Ur Rahman, Ashfaq Mahee Siddiky, Marzan Mahatab, Anik Sinha, and Nova Ahmed. 2024. A Personalized Communication Application for Children with Autism through a Participatory Design Approach. In *Companion of the 2024 Computer-Supported Cooperative Work and Social Computing (CSCW Companion '24)*, November 9–13, 2024, San Jose, Costa Rica. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3678884.3681900>

1 Introduction

Autism spectrum disorder (ASD) is a global complication that also has its roots in Low Middle Income Countries (LMIC) like Bangladesh [7]. Children with Autism (CWA) have difficulties in understanding, thinking, and cognitive skills and lack control over communication, whether it be verbal or nonverbal means [20]. Parents of CWA often suffer from stigma that is driven by a lack of knowledge about ASD in society, cultural beliefs, and a tendency to blame themselves for their child's diagnosis [10]. A lack of infrastructure and assistive technology, along with various barriers, are faced by the parents when ensuring special education for their CWA due to the social stigma associated with the disability [22]. Besides, CWA has a strong affinity with technology, so the development of a technology-based application is required [15].

According to Muneeb et al, applications were favored over non-application-related activities. They did an in-depth analysis that displayed significant improvement in social interaction and communication of CWA after using their application [2]. It is also reported that mobile applications provide a suitable environment for CWA and help to develop their learning and social skills [18, 19]. According to Aguiar et al.'s study, the needs and preferences of CWA should be addressed throughout the design and development of these interventions and by enabling personalization. [1]. For example, to allow to personalize the interface features or the media.

Language learning is also often difficult for CWA, and proper communication plays a vital role in all kinds of learning [18]. As a communication aid to CWA, researchers have created various educational applications that include communication learning or communicating features [18]. Faheem [4], for instance, offers a user-friendly interface with customizable lessons and rewards. It emphasizes the importance of addressing specific learning challenges associated with autism, such as over-selectivity and generalization when developing educational systems for autistic children. In another study, AutiVid [3] is an application designed to aid the CWA in developing daily living skills through video modeling. It prioritizes conducting usability testing to gather comprehensive feedback for app improvement. A subsequent study, Autisay [23], is designed to improve social communication for autistic children by facilitating interaction with caregivers. It suggests that the application should be highly customizable to cater to the individual needs of autistic

children, allowing for the personalization of images, text, and voice. In another study, Nikolay et al. developed a web page reading application, Open Book, for ASD personnel [14]. It emphasizes visual cues, simplified text, and personalization [14]. Researchers such as Nikolay et al and Sampath et al [18] highlight challenges like small picture areas and complex navigation, suggesting solutions such as simplifying the layout, minimizing information overload, and organizing content into distinct categories. Other applications that aid CWA in communicating include Aawn [24], Bolbo Kotha [11], and many more [8, 25].

PECS is a widely used method for communication in CWA through picture exchange [12]. PECS is particularly significant in the context of Autism Spectrum Disorder (ASD), as many individuals with ASD are visual learners and find visual supports like PECS helpful for understanding and communication [13]. Carolyn et al emphasize that the participatory design approach plays an important role in developing a tailored solution that addresses the unique challenges faced by children with autism, leading to better engagement and learning outcomes [21]. Faheem, AutiVid, and Autisay all used PECS, but they did not use it through a participatory design approach in their research method. Ilias et al also highlight the importance of culturally tailored interventions for ASD [10]. However, these applications lack customizability, which makes them unable to be tailored to our cultural context. By cultural context, we mean the ability to tailor the application to our culture, such as incorporating locally available images and making the application accessible to users who speak Bengali. The "Hello" application aids Bengali ASD individuals with PECS [25]. At a certain point, there appear to be some limitations in creating the user's own cards and categories. Although real-life pictures enhance learning for children with ASD [13], the "Hello" application uses illustrated pictures as content for the cards rather than real-life pictures.

Our work is derived from these existing literature gaps-connecting personalization and cultural context using PECS through participatory design. We concentrated on developing a customizable communication application based on cultural context with visual cards, categories, and a simple UI through a participatory design approach. One of our primary objectives was to ensure that the CWA could easily navigate the application and communicate with others. By assisting communication, the application can also aid the CWA to enhance learning and daily living skills.

For the purposes of developing a communication application for CWA, we sat down with three participants multiple times through two participatory design sessions followed by one design workshop attended by designers and researchers. Design Session I started with participants sharing the daily struggles of CWA in communication. During the discussion, a cycle of acknowledgment and appreciation emerged for developing a communication application for CWA. They asserted that the application required to be user-friendly and simple to be effectively used by the CWA. The Design Workshop was attended by designers and researchers to solidify the ideas and insights that were unfolded in Participatory Design Session I. An interactive board and a low-fidelity application design were formulated for Participatory Design Session II. Participants were allowed to redesign the interface of the application in Session II, which aimed for further improvements.

interactive board served as a tangible representation of the application. Pictures used in the interactive board were of household objects to increase familiarity among users. Participatory Design Session II ended with participants' valuable insights regarding existing functionalities and suggestions for new features. The entire procedure from the setup to the study can be seen in Figure 1.

2.3 Data analysis

The research team that led the design sessions also collaborated on translation, transcription, and data analysis. A total of 1.5 hours of Audio was translated and transcribed. Following our design sessions, we went through the transcriptions to identify key themes to guide the development of the application's interface design, functionalities and features. We replicated the procedure to assess user experience, accessibility, learnability, effectiveness, and engagement. The authors conducted several sessions to derive, categorize, and implement the findings from our interviews.

2.4 Ethics and Incentives

This study was approved by the author's institution's Institutional Review Board (IRB). All the participants were informed about the purpose of the study prior to the interview. We provide a comfortable and culturally sensitive environment for the design session. All the participants' data were stored in a safe drive where only the researchers had access. Participants did not receive any incentives for their involvement in the study.

3 Findings

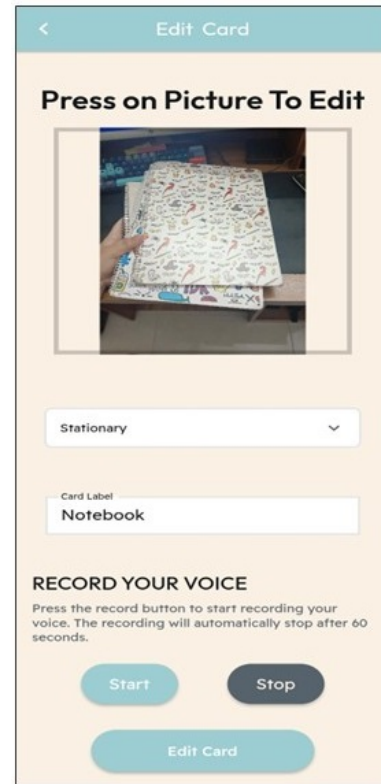
The findings of this study was based on the feedback from the participants P1, P2, and T1 during two sessions, Participatory Design Session I and II. We discussed the initial idea of the features of the future application in the first session, and explored iterative feedback on the developed application in the second session.

3.1 Participatory Design Session I

The focus of this session was to design an application based on the participant's suggested features. The features had to be accessible, intuitive, and simple for the CWA.

Feature: We decided to design a card-based application based on the participant's suggestion. The cards will contain a picture, name, and pre-recorded audio that will play on selection. They suggested adding categories to organize the cards and improve accessibility. They emphasized personalizing the cards and categories. Figure 3 shows how an existing card can be personalized by a user. They wanted to sync the data for cross-device usage.

Accessibility: The parents suggested limiting the access of the application to one page since having to navigate multiple pages can be overwhelming for the CWA. A blank canvas will be at the top followed by cards in the middle and categories at the bottom. Selecting the cards will add them to the canvas above. It will have a play and remove button to play the cards or clear the canvas. The cards and categories would each be in three columns. They suggested a hamburger icon at the top left that would open a sidebar with buttons for the cards, categories, and account pages. Figure 4 (a), displays the procedure of creating a Category by tapping on the "Add" button represented by '+' symbol, available in the Categories



Page layout of the card editing page

Figure 3: Exhibits the scratch version of our proposed idea of the application, which is the low-fidelity design of the application. This figure shows the state where we designed and defined our application's different elements, functionalities and page layouts. Rather than technicalities, solidifying our idea was the main purpose of this task.

page. Figure 4 (b), shows the process of creating a Card using the "Add" button option, represented by '+' symbol, available in the Cards page.

Personalization: On the cards and categories page, there will be an 'Add' button to create them. The user will add a picture, name, category, and pre-recorded audio. The category will have a picture and a name. All the cards and categories will be shown on their respective pages. Tapping on existing cards or categories will enable editing.

3.2 Participatory Design Session II

After the development of the application, we presented it to the same study participants as Participatory Design Session I. Along with the application, our team prepared an interactive board of the application with cutouts of the page elements. The participants were asked to liberally re-design the interface to their preferences. We received feedback based on the interactive board and the developed

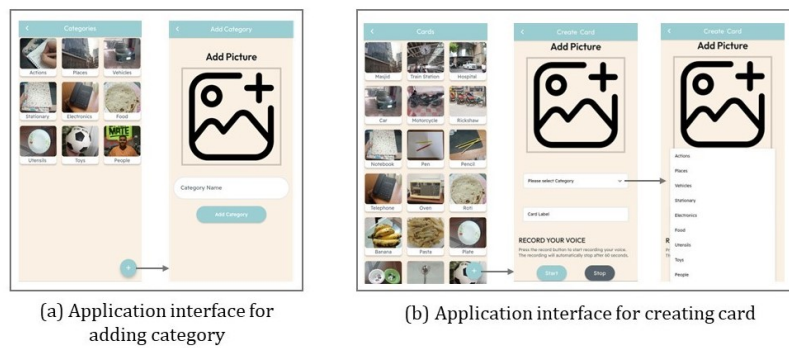


Figure 4: (a,b) Resemble the application interface for creating a Category and Card respectively.

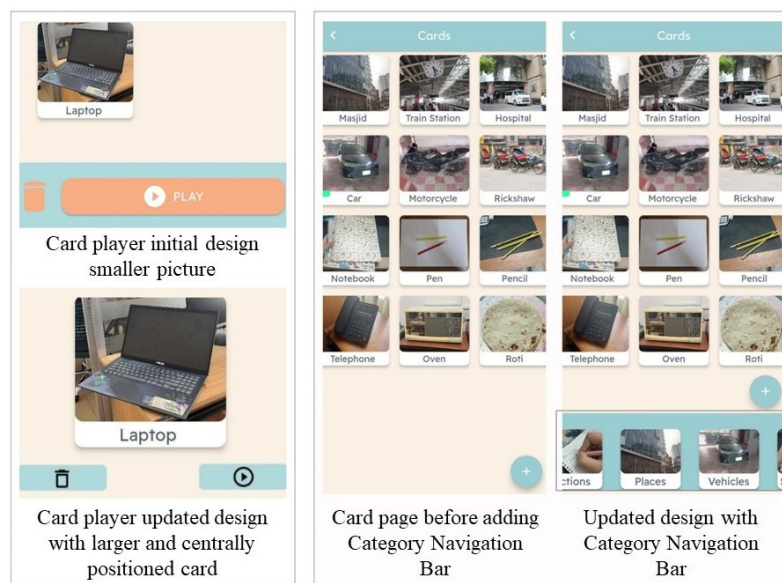


Figure 5: Updated Application Interface

application. The participants showed enthusiasm in the design process.

3.2.1 Application Features. According to the suggestions by the participants, the main interface of the application focused on being intuitive and simple for CWA's efficient communication. We incorporated the participants' suggestions and sought their feedback to develop the application.

Accessibility: Participants gave suggestions to make the application more accessible, like immediate audio playback upon card selection - removing extra steps and reducing frustration for CWA. For the visual comfort of the CWA, they also suggested larger and centrally positioned cards in the card player for visual comfort, suggesting maximum 3 cards at a time and dynamic layout adjustment for newly added cards. The participants praised the sixcard layout below the card player as it reduced clutter and made the interface look clean. Keeping the minimalist interface in mind, calming colors were selected for the application interface and absence of

loud noises or sudden effects garnered positive feedback from the parents and teachers.

Personalization: One of the main focuses was to create a significantly customizable application, as each CWA adjusts differently, according to participants. The user can make cards with words or full sentences. The user will manually add the audio. According to the participants, since CWA are visual learners, the cards will have pictures to help the child understand better. A parent mentioned their child's visual learning and the teacher agreed.

"Most autistic children, including my child, are visual learners. Most of them. This is not hard for them. If there's 100/200 pictures, he can understand very quickly." – Parent

It is important that an application created for the benefit of CWA be customizable according to their comfort. Hence, the customization feature was appreciated by the participants.

3.2.2 Application-facilitated Additional Advantages. Efficiency: The participants highlighted the importance of the efficiency of the application. They wanted a limited number of taps/clicks for navigating different pages. According to the teacher T1, this is to avoid the child getting frustrated from repetition. The participants also wanted a category panel on the landing page, so that the children could swiftly find the cards they need. Tapping the card will play the recordings. Tapping the card again will remove it from the card player. The participants mentioned, returning to the landing page from anywhere will be easier with a 'back' button or a 'Category Navigation Bar' at the bottom of the screen. This will allow the children to easily navigate between the categories without getting frustrated. The participants found these features to be user-friendly and efficient for the CWA. Figure 5 shows an updated interface developed after the feedback taken from the participants.

Aid in Learning: The teacher participant present in the study mentioned how the application can be used in learning. If a nonverbal child wanted to learn something, they could easily communicate it to their teacher using the application. CWA are also visual learners, so the addition of pictures with the cards can help them grasp things quickly. According to the participants, if used in a regulated and controlled environment, this application can be an excellent tool of education for CWA.

4 Conclusion and Way Forward

Our work recognizes the existing literature gap and provides a customizable mobile communication application in the South Asian context. We conducted a participatory design approach while developing our application to address the unique communication challenges faced by CWA. It also ensured the design and features of our application were directly informed by the needs and experiences of the community. The visual card feature of our application can not only assist CWA in communication but also has the potential to assist the education of CWA. However, it is also important to acknowledge the limitations of our study, particularly the small sample size. Moving forward, we will conduct further studies, including evaluating the application by deploying it to the parents and teachers of CWA. We also intend to collaborate directly with CWA for additional application evaluation once it reaches the ideal phase. We believe our proposed solution will make communication easier for CWA. Through our solution, we envision a future where CWA can communicate with more clarity; especially in South Asian countries.

References

- [1] Yuska Paola Costa Aguiar, Edith Galy, Anaïs Godde, Maëla Trémaud, and Carole Tardif. 2022. AutismGuide: a usability guidelines to design software solutions for users with autism spectrum disorder. *Behaviour & Information Technology* 41, 6 (2022), 1132–1150.
- [2] Muneeb Imtiaz Ahmad and Suleman Shahid. 2015. Design and evaluation of mobile learning applications for autistic children in Pakistan. In *Human-Computer Interaction—INTERACT 2015: 15th IFIP TC 13 International Conference, Bamberg, Germany, September 14–18, 2015, Proceedings, Part I* 15. Springer, 436–444.
- [3] Wan Fatimah WAN AHMAD and Kavisha A/P SATIAKUMAR. 2022. Autivid: A Mobile Application to Improve Daily Living Skills of Autistic Children through Video Modeling. *Asian Journal of Research in Education and Social Sciences* 4, 1 (2022), 284–291. <https://doi.org/10.55057/ajress.2022.4.1.24>
- [4] Saleh Alhazbi, Amr Aboelenen, Naheel Faisal Kamal, Mohammad Khader, Adel Azzouza, Ayman Al-Kababji, Abdulahi Hassen, and Ahmad Zaza. 2018. Faheem: A Tablet-Based Application to Improve Receptive Language for Arab Autistic Children. 248–252. <https://doi.org/10.1109/COMAPP.2018.8460413>
- [5] S. Bødker, C. Dindler, O.S. Iversen, and R.C. Smith. 2022. *What Is Participatory Design?* Springer, Cham. https://doi.org/10.1007/978-3-031-02235-7_2
- [6] S. Bødker, C. Dindler, O. S. Iversen, and R. C. Smith. 2016. Unpacking the Notion of Participation in Participatory Design. *Computer Supported Cooperative Work (CSCW)* 25, 6 (2016), 425–475. <https://doi.org/10.1007/s10606-016-9259-4>
- [7] Gauri Divan, Supriya Bhavnani, Kathy Leadbitter, Ceri Ellis, Jayashree Dasgupta, Amina Abubakar, Mayada Elsabbagh, Syed Usman Hamdani, Chiara Servili, Vikram Patel, et al. 2021. Annual Research Review: Achieving universal health coverage for young children with autism spectrum disorder in low- and middle-income countries: A review of reviews. *Journal of Child Psychology and Psychiatry* 62, 5 (2021), 514–535.
- [8] Samir El-Seoud, AbdelGhani Karkar, Jihad Aljaam, and Omar Karam. 2014. A Pictorial Mobile-based Communication Application for Non-Verbal People with Autism. In *Proceedings of 2014 International Conference on Interactive Collaborative Learning, ICL 2014*. <https://doi.org/10.1109/ICL.2014.7017828>
- [9] Anurata Prabha Hridi, Anik Saha, Anik Sinha, Ifiti Azad Abeer, Nova Ahmed, and Shameem Ahmed. 2021. Understanding the Education of Children with Autism in Bangladesh: Parents' Perspective. *Proc. ACM Hum.-Comput. Interact.* 5, CSCW1 (2021), Article 60. <https://doi.org/10.1145/3449134>
- [10] Kartini Ilias, Kim Cornish, Auretta S Kummar, Miriam Sang-Ah Park, and Karen J Golden. 2018. Parenting stress and resilience in parents of children with autism spectrum disorder (ASD) in Southeast Asia: A systematic review. *Frontiers in psychology* 9 (2018), 280.
- [11] Mohammad Nasfikur Khan, Faisal Shahin, Fahim Sunny, Moshir Khan, A. K. Mashuk, and Khondaker Mamun. 2019. An Innovative and Augmentative Android Application for Enhancing Mediated Communication of Verbally Disabled People. 1–5. <https://doi.org/10.1109/ICCCNT45670.2019.8944655>
- [12] J. G. C. Lendio, S. M. C. Teves, and D. S. Catalan. 2023. Picture Exchange Communication System and Its Effectivity to the Communication of Learners with Autism Spectrum. *Cognizance Journal* 3, 3 (2023), 90–304. <https://doi.org/10.47760/cognizance.2023.v03i03.007>
- [13] M. Mohamad, P. N. Esa, and R. Kamarudin. 2022. Parents' Perceptions of Using Picture Exchange Communication System (PECS) in English for Autism Spectrum Disorder (ASD) Children. *Creative Education* 13, 11 (2022), 3602–3623.
- [14] Nikolay Pavlov. 2014. User Interface for People with Autism Spectrum Disorders. *Journal of Software Engineering and Applications* 7 (2014), 128–134. <https://doi.org/10.4236/jsea.2014.72014>
- [15] Cynthia Putnam and Lorna Chong. 2008. Software and technologies designed for people with autism: what do users want? In *Proceedings of the 10th international ACM SIGACCESS conference on Computers and accessibility*. 3–10.
- [16] Maun Rachael, Fabri Marc, and Trevorrow Philippa. 2023. Participatory Methods to Engage Autistic People in the Design of Digital Technology: A Systematic Literature Review. *Journal of Autism and Developmental Disorders* (2023). <https://doi.org/10.1007/s10803-023-06015-5>
- [17] Skovbo Rasmussen Pernille and Katrine Pagsberg Anne. [n. d.]. Customizing Methodological Approaches in Qualitative Research on Vulnerable Children with Autism Spectrum Disorders. ([n. d.]). <https://doi.org/10.3390/SOC9040075> [n. d.].
- [18] Harini Sampath, Ravi Agarwal, and Bipin Indurkha. 2013. Assistive technology for children with autism-lessons for interaction design. In *Proceedings of the 11th Asia Pacific conference on computer human interaction*. 325–333.
- [19] Harini Sampath, Bipin Indurkha, and Jayanthi Sivaswamy. 2012. A communication system on smart phones and tablets for non-verbal children with autism. In *Computers Helping People with Special Needs: 13th International Conference, ICCHP 2012, Linz, Austria, July 11–13, 2012, Proceedings, Part II* 13. Springer, 323–330.
- [20] Sayma Alam Suha, Muhammad Nazrul Islam, Shammil Akter, Milton Chandro Bhowmick, and Rathin Halder. 2022. Assessing usability of mobile applications developed for autistic users through heuristic and semiotic evaluation. In *Proceedings of International Joint Conference on Advances in Computational Intelligence: IJCAI 2021*. Springer, 25–39.
- [21] C. Ten Holter. 2022. Participatory design: lessons and directions for responsible research and innovation. *Journal of Responsible Innovation* 9, 2 (2022), 275–290.
- [22] Tanjina Farhana Upoma, Mst Sabrina Moonajilin, Md Estiar Rahman, and Most Zannatul Ferdous. 2020. Mothers initial challenges having children with autism spectrum disorders in Bangladesh. *Bangladesh Journal of Medical Science* 19, 2 (2020), 268–272.
- [23] SitiNor Baziliah Nyuk Voon, Abdullah Maidin, Halina Jumaat, and MuhammadZulfadhl Ahmad. [n. d.]. *AutiSay: A Mobile Communication Tool for Autistic Individuals*. https://doi.org/10.1007/978-3-319-13153-5_34 [n. d.].
- [24] Arwa Wali, Modi Alfrhidi, Nada Alasiri, and Najah Alsabei. 2023. Aawn: An Interactive Mobile Application for Improving the Communication Skills of Arab Children with Autism. *TEM Journal* 12 (2023), 1307–1315. <https://doi.org/10.18421/TEM123-10>
- [25] Ishrat Yra, Maksud Pranto, and Amit Das. 2022. HELLO: An Android-based Mobile Application for Autism Individuals to Improve Socio-Communicational Learnability in Bangladesh. 1–7. <https://doi.org/10.1109/IMCOM53663.2022.9721813>